

APM1110-FA6_Paglinawan_ZeusAlexis

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I. Geometric Distribution

Provide an R code for the geometric distribution. The geometric distribution is a probability distribution that models the number of trials required to achieve the first success in a sequence of Bernoulli trials, where each trial has a constant probability of success.

1. Set the probability of success: $p \leftarrow 0.2$
2. Generate 1000 random variables from the geometric distribution.
3. Calculate basic statistics:

```
mean_x <- mean(x)
```

```
var_x <- var(x)
```

```
sd_x <- sd(x)
```

4. Print the results in item 3 with the following output format (string):

Number of trials required to achieve first success:

Mean (in 2 decimal places):

Variance (in 2 decimal places):

Standard deviation (in 2 decimal places):

5. Plot the histogram of the results.

```
# 1 Probability of success
p <- 0.2

# 2 Generate 1000 random variables
x <- rgeom(1000, p) + 1

# 3 Calculate basic statistics
mean_x <- mean(x)
var_x <- var(x)
sd_x <- sd(x)

# 4 Print results
cat("Number of trials required to achieve first success: \n")
```

Number of trials required to achieve first success:

```

for (i in seq(1, length(x), by = 20)) {
  cat(paste(x[i:min(i + 19, length(x))], collapse = " "))
  cat("\n")
}

```

```

8 17 1 4 7 1 1 4 7 3 1 1 1 1 1 2 1 4 4 6
7 10 3 32 5 8 7 13 5 15 8 3 7 2 6 5 6 1 3 2
4 2 11 13 9 5 3 6 3 13 2 16 4 5 2 1 7 3 2 5
3 5 4 1 1 4 1 2 2 7 7 1 3 3 8 2 3 4 1 7
3 2 11 1 6 4 3 3 4 2 3 1 9 4 11 7 18 1 8 3
13 6 1 1 5 2 1 1 1 8 2 3 7 3 2 2 6 8 3 2
8 2 14 9 2 1 4 3 19 4 2 9 7 2 4 1 12 2 7 3
1 9 6 1 20 1 7 1 6 1 1 3 19 2 5 6 3 7 1 6
3 3 4 5 17 2 6 2 1 1 5 10 1 1 11 4 3 2 1 12
4 2 3 4 2 4 8 3 4 3 5 8 6 5 9 4 1 5 7 1
2 5 5 3 1 4 4 9 1 3 10 12 11 2 7 1 4 1 1 5
1 5 8 2 1 3 11 2 22 6 4 3 3 2 11 2 15 2 3 19
5 6 8 4 2 3 2 14 6 2 9 8 14 5 4 3 5 12 2 19
1 2 1 1 1 1 6 2 6 2 5 1 9 2 8 1 1 3 1 12
4 9 5 4 2 1 3 7 7 1 21 5 1 1 1 16 3 9 2 4
3 2 1 1 9 2 2 7 5 11 7 3 7 6 4 2 14 1 8 3
3 4 12 1 1 7 2 2 2 20 4 9 10 1 11 12 5 3 3 6
6 2 4 5 2 1 1 5 6 4 10 9 8 18 2 2 3 5 9 18
6 14 3 7 3 8 11 1 6 1 4 5 3 6 1 3 1 3 1 1
2 3 2 6 3 2 3 1 7 3 3 1 3 7 3 3 5 9 5 2
3 4 3 2 2 7 1 8 4 1 4 6 6 2 2 11 13 3 2 1
8 3 16 6 2 13 4 1 1 3 4 1 2 2 6 3 6 6 7 1
9 2 8 3 4 2 4 5 1 32 9 1 3 1 5 9 3 1 3 8
3 7 1 2 2 7 3 4 5 5 6 3 3 7 1 8 1 9 3 3
2 2 1 1 5 7 5 2 3 9 2 4 2 2 5 4 1 1 9 1
5 2 8 2 2 3 2 9 5 2 8 2 1 1 1 8 3 2 2 18
10 5 14 4 4 2 3 2 5 2 3 3 9 1 1 18 11 6 5 4
12 5 2 8 9 6 7 2 1 1 1 9 18 2 5 5 13 16 1 1
8 1 1 6 4 3 4 9 4 4 11 3 3 3 9 1 2 1 3 4
1 4 1 9 4 8 17 2 3 2 1 3 4 6 6 6 2 1 10 7
10 6 2 1 6 2 3 4 8 3 4 3 3 1 3 6 1 3 9 2
6 5 4 6 6 1 2 2 7 9 3 4 4 4 1 1 18 1 6 4
4 3 4 14 1 4 3 13 5 2 7 1 2 1 4 7 10 1 8 6
4 4 9 3 23 4 2 12 4 5 12 8 7 13 3 1 5 2 3 2
1 6 3 11 10 7 11 3 6 11 5 1 10 21 1 1 2 16 4 1
2 4 11 3 5 4 1 19 1 2 8 16 9 1 6 10 2 9 19 6
1 4 4 1 9 1 6 5 13 4 2 3 14 4 2 2 2 8 3 5
1 6 1 1 1 4 1 11 5 3 1 8 1 4 4 3 2 1 1 2
17 1 2 4 4 6 3 7 4 2 2 1 4 4 13 1 3 3 20 2
4 3 3 12 5 3 3 4 1 2 4 2 2 3 6 13 11 5 2 5
7 4 2 1 9 2 2 6 12 4 12 2 1 8 3 11 4 3 1 2
1 7 5 4 1 2 7 7 2 6 8 1 1 6 3 2 2 3 2 2
6 4 5 25 5 7 2 2 16 2 7 2 19 16 1 1 6 2 9 3
2 6 2 3 7 16 4 18 2 4 10 26 1 7 2 1 5 13 1 2
4 4 1 2 5 2 3 14 8 12 5 9 2 2 4 1 1 4 2 2
2 2 3 3 7 10 25 2 2 4 22 5 1 7 4 4 19 7 11 1
1 4 1 3 4 5 2 3 1 1 18 5 11 4 5 4 8 2 8 4
2 7 6 2 8 5 2 3 1 6 1 8 3 2 4 2 11 16 3 1

```

```
2 5 2 2 2 1 1 3 2 2 6 5 13 1 3 2 9 19 2 8
4 7 8 5 11 2 2 2 2 1 7 2 4 11 7 2 9 14 13 4
```

```
cat("Mean (in 2 decimal places):", sprintf("%.2f", mean_x), "\n")
```

Mean (in 2 decimal places): 5.04

```
cat("Variance (in 2 decimal places):", sprintf("%.2f", var_x), "\n")
```

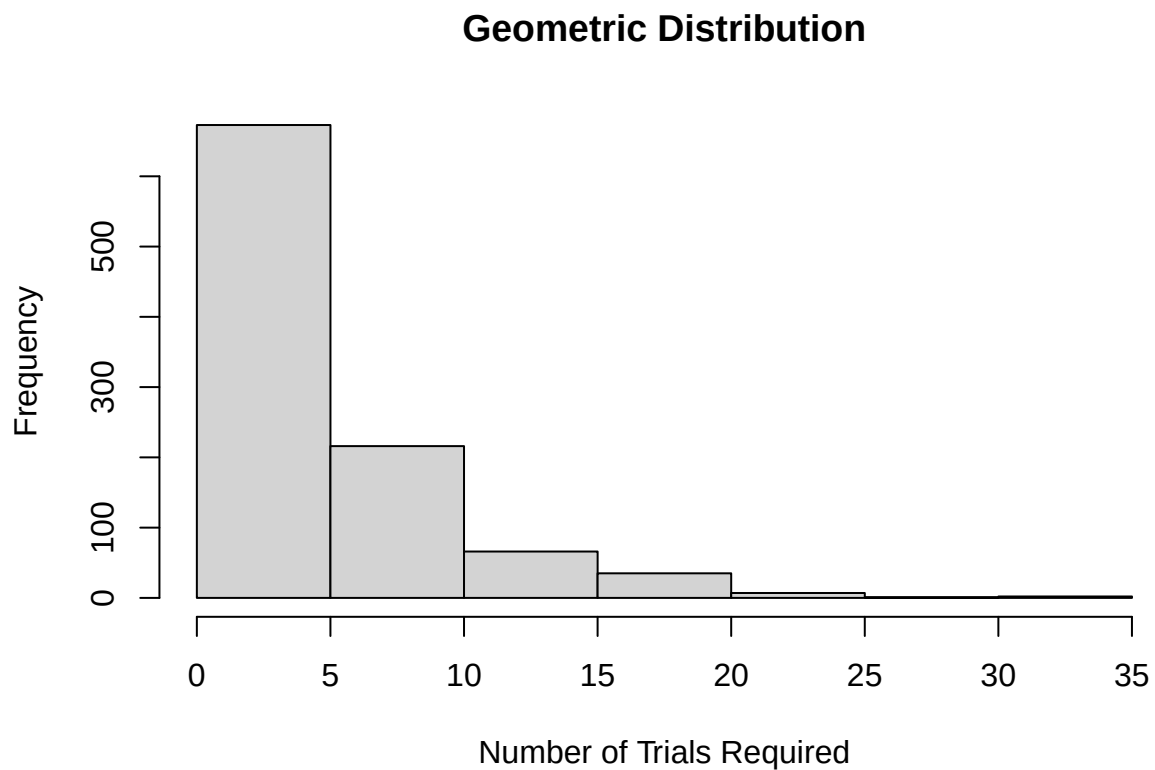
Variance (in 2 decimal places): 20.17

```
cat("Standard deviation (in 2 decimal places):", sprintf("%.2f", sd_x), "\n")
```

Standard deviation (in 2 decimal places): 4.49

```
# 5 Histogram
```

```
hist(x,
     main = "Geometric Distribution",
     xlab = "Number of Trials Required",
     ylab = "Frequency")
```



II. Hypergeometric Distribution

Consider a plant manufacturing IC chips of which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box. Estimate the probability that the sample contains more than 10% defectives, when:

1. A sample of 10 is selected from a box of 40;
2. A sample of 10 is selected from a box of 5000.

```
# Box of 40
box1 <- 40
defective1 <- 4
sample1 <- 10

# More than 10% of 10 means at least 2 defective chips
P1 <- 1 - phyper(1, defective1, box1 - defective1, sample1)

cat("Probability for a sample of 10 from a box of 40:",
    sprintf("%.4f", P1), "\n")
```

Probability for a sample of 10 from a box of 40: 0.2559

```
# Box of 5000
box2 <- 5000
defective2 <- 500
sample2 <- 10

# More than 10% of 10 means at least 2 defective chips
P2 <- 1 - phyper(1, defective2, box2 - defective2, sample2)

cat("Probability for a sample of 10 from a box of 5000:",
    sprintf("%.4f", P2), "\n")
```

Probability for a sample of 10 from a box of 5000: 0.2639

Interpretation

For a box of 40 chips, 10% defective means there are 4 defective chips. Since the sample size is 10, more than 10% defective means having at least 2 defective chips. The probability of this occurring is 0.2559, or 25.59%.

For a box of 5000 chips, 10% defective means there are 500 defective chips. Again, more than 10% defective in a sample of 10 means at least 2 defective chips. The probability of this occurring is 0.2639, or 26.39%.